

RC-NFGD | reliability-gated force feedback

1,000-step reverse diffusion

window enlarged for clarity

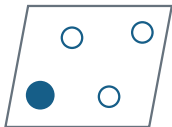
last 20 predictors only

$t = 1$

$t = 0.02$

ϵ

01 CLEAN STRUCTURE



predicted clean crystal
from current MatterGen state

02 FORCE FEEDBACK



frozen MatterSim-5M
center + bound Cartesian force

03 SCORE UPDATE

Cartesian $\rightarrow$ fractional

$$\delta f = \delta r H_{\vec{0}}^{-1}$$

position predictor only

$$\tilde{s}_t^f = s_t^f + \delta f / c_t^f$$

atomic / cell / corrector unchanged



ACCEPT finite output, valid cell / score coefficient, and distance ≥ 0.5 Å

Otherwise keep the original MatterGen score; no feedback is applied.

$$\eta = 0.005 \text{ Å} \cdot \text{hard cap} = 0.01 \text{ Å} \cdot \text{CFG} = 2.0$$